

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION
Detailed Marking Report — Physics Exam-15

Seemab · Class 9 (Science Group) · Units 1 to 6 (Unit 6 to §6.4) and Unit 9 · Attempted 10 September 2026 · deep-check verified

45 / 46
TOTAL

97.8%
PERCENTAGE

A+
GRADE

29 / 30
SECTION A

8 / 8
SECTION B

8 / 8
SECTION C

SECTION A — every question (29 / 30)

#	Question	Hers	Key	Mark	Note and source in her textbook
1	The number of significant figures in 0.02080 kg is:	C	C	1	U1 §1.10, P-09-A-16 (rounding/significant-figure conventions). Source: U1 p.30–31, general rules for writing significant figures.
2	Arrows land in a tight cluster, far from the bullseye:	B	B	1	U1 §1.9, P-11-A-09 (differentiate precision and accuracy).
3	Instrument to measure WEIGHT directly:	A	B	0	Her written reason names the spring balance (B) but she recorded A. Written letter governs: wrong. U1 §1.7.4 (physical balance); U3 P-09-B-22 (force meter measures weight); U5 §5.4.1 (spring scale).
4	Water rises from 42 mL to 65 mL:	A	A	1	U1 §1.7, P-09-A-13 (measuring cylinder to measure volume).
5	Ball dropped, reaches 35 m/s:	C	C	1	U2, P-09-B-10 (free-fall acceleration to solve problems).
6	Horizontal line on a distance-time graph means:	A	A	1	U2, P-09-B-9 / P-09-B-11 (sketch/interpret distance-time graphs; gradient gives speed).
7	Train's speed 10→40 m/s in 15 s, GRADIENT represents:	A	A	1	U2, P-09-B-14 / P-09-B-15 (gradient of speed-time graph gives acceleration).
8	Car at constant 12 m/s for 25 s, AREA represents:	B	B	1	U2, P-09-B-12 / P-09-B-13 (area beneath speed-time graph gives distance).
9	Trolley 0.5 kg, 20 m/s, stopped in 0.05 s. Impulse:	D	D	1	U3 §3.6, P-09-B-40 (define and calculate impulse).
10	Bullet 0.01 kg/200 m/s embeds in 1.99 kg block:	A	A	1	U3, P-09-B-44 (apply conservation of momentum in one dimension).
11	F = 18 N produces a = 2 m/s². Mass:	C	C	1	U3, P-09-B-30 (state and apply Newton's second law).
12	Table 3.1, proton-electron interaction is an example of:	B	B	1	U3 §3.1.2, Table 3.1; P-09-B-25 (three/four fundamental forces and their relative strengths).
13	Seesaw: 30 N at 2 m (left); balancing weight at 3 m (right)?	A	A	1	U4, P-09-B-45 / P-09-B-46 (equilibrium using the principle of moments).
14	Four engines, all thrust in the same direction:	D	D	1	U4 §4.1, P-09-B-43 (differentiate like and unlike parallel forces).
15	Skydiver reaches terminal velocity when:	C	C	1	U4, P-09-B-36 (analyse the dynamics of an object reaching terminal velocity).
16	Disadvantage of friction:	B	B	1	U4 §4.6.2 (advantages and disadvantages of friction).
17	Well-type manometer, $\rho=1200\text{ kg/m}^3$, $h=15\text{ cm}$. Pressure difference:	D	D	1	U5 §5.5, P-09-B-85 (workings and applications of a manometer).
18	4 kg hung from spring, extends 2 cm. Spring constant:	no letter	C	1	No letter written; working gives 2000 N/m = option C. Credited (working governs only when no letter exists). U5 §5.2, P-09-B-56 (define and calculate the spring constant).
19	One pascal is defined as a pressure of:	A	A	1	U5, P-09-B-77 (define and calculate pressure).
20	Hydraulic press: $A_1=0.0015\text{ m}^2$, $A_2=0.45\text{ m}^2$, $F_1=20\text{ N}$. Output force:	D	D	1	U5 §5.6, P-09-B-86 (define and apply Pascal's law).
21	Three common types of manometer:	B	B	1	U5 §5.5, P-09-B-85.
22	4 kg ball dropped from 10 m. Kinetic energy at impact:	C	C	1	U6, P-09-B-64 / P-09-B-65 / P-09-B-68 (KE, PE, conservation of energy).
23	Sledge pulled with 90 N at 60°, moved 3 m. Work done:	A	A	1	U6 §6.1 (Vector Representation, Fig. 6.2), P-09-B-61 ($W = Fd$, force at an angle).
24	Renewable energy resource:	D	D	1	U6 §6.3.6, P-09-B-70 (differentiate and list renewable/non-renewable resources).
25	Motor lifts 200 kg through 6 m in 10 s. Power:	C	C	1	U6, P-09-B-73 (define and calculate power).
26	Fuel used in nuclear power plants:	B	B	1	U6 §6.3.2 Nuclear Fuel, P-09-B-71 / P-09-B-72.
27	Physics textbook's Key Points define technology as:	D	D	1	U9 §9.4, P-09-G-08 (differentiate science, technology, engineering).
28	Branch of physics studying Earth's climate:	A	A	1	U9 §9.2 (branches of physics), P-09-G-02.
29	Best definition of a 'hypothesis':	C	C	1	U9 §9.3, P-09-G-06 (understand hypothesis, theory and law).
30	Physics is a branch of:	D	D	1	U9 introduction; Exercise Q8; P-09-G-03 (physics as a subset of physical/natural sciences).

Red rows lost the mark; amber rows are judgement calls. Every letter was re-read from the scan and cross-checked by an independent deep-check pass.

45

out of 46 · 97.8%

29/30

MCQs · 97%

16/16

Sections B + C

The short version. Seemab scored 45 out of 46. This is her strongest paper of the year, and the reason matters more than the number.

Three weeks ago her multiple choice was the clear weak point of her work. On Maths exam-17 it was 60%. On Chemistry exam-21 it was 83%. On this paper it is **97%**. She lost exactly one mark on Section A, and even that one she knew the answer to.

The change is visible on the page. This is the first attempt where she has written out working and a stated reason for her multiple choice answers instead of just letters. Question 1 explains why the leading zeros in 0.02080 do not count. Question 6 explains why the body is at rest rather than accelerating. Question 16 explains why the odd one out is a disadvantage rather than an advantage. That was the single piece of advice given to her, and she acted on it.

All four repeat errors: corrected. Four questions on this paper were chosen deliberately because they are the exact multiple-choice questions she has got wrong on earlier papers. She answered all four correctly, three of them with working shown.

Q	Her old mistake	This time
7	Gradient of a speed-time graph, wrong on exam-12 and again on exam-14	$a = (40-10)/15 = \mathbf{2\ m/s^2}$ ✓
8	Area under a speed-time graph, her longest-standing physics error	distance = $12 \times 25 = \mathbf{300\ m}$ ✓
19	The SI unit of pressure, wrong on exam-12	$\mathbf{1\ N/m^2}$ ✓
24	Renewable vs non-renewable, wrong on exam-14	Geothermal ✓

Gradient against area was the pair she had never separated. She got both directions right, in the same paper, with the arithmetic written out.

The one mark lost, and it is a frustrating one

Question 3 asked which instrument measures weight directly. The options were A physical balance, B spring balance, C measuring cylinder, D vernier caliper. **She wrote A.**

But her own written reason underneath says, word for word: *"because spring balance is measuring force, measuring cylinder is used to measure volume and vernier calipers are used to measure small lengths."*

She identified the spring balance as the force-measuring instrument and correctly ruled out two of the other three. She reasoned her way to B and then recorded A. **The mark is lost to transcription, not to knowledge.**

This is worth saying to her directly, because it is now the second paper running where a mark went this way: on Maths exam-17 she wrote a coordinate instead of an option letter and it could not be read. **The habit to build is simple: after writing the reason, look back at the options and check the letter you have written actually matches the one you argued for.**

Section A — 29/30

Twenty-eight letters correct, one wrong (Q3), and one question answered by working alone (see below). A shift guard was run to rule out a mis-numbered answer list: 29 matches at the correct alignment against 1, 1, 3 and 1 at the four neighbouring offsets. No misalignment.

Q	Topic	Her working, quoted
1	Significant figures	"ignoring first two zeros we have 2080"
5	Free fall	$t = (35 - 0)/10 = 3.5\ s$
9	Impulse	$I = (0.5)(0) - (0.5)(20) = -10\ N\cdot s$
10	Conservation of momentum	$0.01 \times 200 + 1.99 \times 0 = 2$, so $v = 1\ m/s$
13	Principle of moments	$30 \times 2 = F_2 \times 3$, $F_2 = 20\ N$
17	Manometer	$P = \rho gh = 1200 \times 10 \times 0.15 = 1800\ Pa$
20	Hydraulic press	$F_2 = 20 \times 0.45 / 0.0015 = 6000\ N$
22	KE and PE conservation	$KE = PE = mgh = 4 \times 10 \times 10 = 400\ J$
23	Work at an angle	$W = 90 \times 3 \times \cos 60^\circ = 135\ J$

Question 18 needs a decision. She wrote **no option letter** at all, only the working: $F = kx$, $k = 40 / 0.02 = \mathbf{2000\ N/m}$. That value is correct and matches exactly one option, C.

I have **awarded** the mark, on the rule that a written letter is the authoritative answer when one exists, and the working governs only when no letter was written. Under that same rule Q3 stays wrong, because there she did write a letter.

Note the precedent: on Maths exam-17 Q10 she also wrote a value instead of a letter and it was ruled wrong. The difference relied on here is that Q10 was genuinely illegible, while this is unambiguous and correct. **If you would rather be consistent with that ruling, the paper is 44/46 (95.7%).**

Section B — 8/8

Q	Topic	Mark	Comment
2(i)	Theory vs law	4/4	Full comparison table. Theory explains <i>why</i> and can be modified with new evidence; law describes <i>what</i> happens, often mathematically. Examples: Cell Theory and Atomic Theory against Hooke's Law and Newton's Law of Universal Gravitation. Her examples differ from the key's suggestions and are equally valid.
2(ii)	Weight on Earth and Moon	4/4	(a) $55 \times 9.8 = 539\ N$. (b) $55 \times 1.62 = 89.1\ N$. (c) <i>"the mass does not change because mass is the amount of matter in an object and remains constant regardless of the gravitational field. Only the weight changes because the moon has a weaker gravitational field than earth."</i> Complete, with the reason stated.

Both were items her school examined that had never appeared on any of our fourteen previous physics papers.

Section C — 8/8

The full satellite orbital numerical, which the coverage tracker had flagged as the longest-standing gap in Unit 4.

Part	Required	Her answer	Mark
(a)	Define orbital motion, state the speed formula	Correct, $v = 2\pi r/T$	2/2
(b)	Orbital speed in m/s	7851 m/s	2/2
(c)	Convert to km/h	28,264 km/h	1/1
(d)	Name the force and its direction	Gravitational, towards the centre of the Earth	1/1
(e)	Centripetal force on a 750 kg satellite	$6.80 \times 10^3 \text{ N}$	2/2

On (b) the exact value is 7854 m/s. Her 7851 is a small unshown-work slip in the long division, about 0.05% out; no standard value of π produces 7851 exactly. The method and setup are perfect and the deviation is immaterial, so full marks stand. Her (c) and (e) were then checked against *her own* figure rather than the ideal one: $7851 \times 3.6 = 28,264$, and $750 \times 7851^2 / 6,800,000 = 6798 \text{ N}$, which rounds to the $6.80 \times 10^3 \text{ N}$ she wrote. Both are genuinely consistent.

What to say to her

- **The rough working worked.** 60% to 83% to 97% in four days, on the one thing she was asked to change. She should hear that plainly.
- **Gradient and area are fixed.** The error she had repeated across two papers did not appear, in either direction.
- **Check the letter against the reason.** The single lost mark was a question she had already reasoned correctly. Ten seconds of checking would have made this a perfect paper.
- **Always write the option letter,** even when the working makes the answer obvious. On Q18 it happened to be recoverable. It has not always been.

Marking independently deep-checked: all 30 MCQ answers re-derived from the printed question stems rather than the answer key (no key error found), her answers re-read letter by letter, a shift guard run across five alignments, and every calculation in Sections B and C re-worked by hand. Both passes reached 45/46 with no disagreement on any question.